



09 — Working Memory Gating (Context)

Overview

The working memory gate is the bottleneck between long-term memory retrieval and prompt context insertion. It scores activated schemas on query relevance, scope match, class/layer bonuses, and noise penalties, then applies MMR deduplication and a token budget.

Mathematical Formulation

Step 1: Schema Scoring

For each schema s with embedding $\mathbf{e}_s$ and cue c :

$$\text{score}(s, c) = \underbrace{\cos(\mathbf{e}_q, \mathbf{e}_s)}_{\text{cosine}} + \underbrace{B_{\text{identity}}(s, c)}_{\text{identity bonus}} - \underbrace{P_{\text{noise}}(s)}_{\text{noise penalty}}$$

Identity bonus B_{identity} is the sum of:

Bonus	Value	Condition
<code>class_bonus</code>	<code>+0.05</code>	Schema type is in <code>allowed_classes</code>
<code>layer_bonus</code>	<code>+0.03</code>	Memory layer is in <code>allowed_memory_layers</code>
<code>provenance_bonus</code>	<code>+0.05</code>	Source kind is <code>"explicit_remember"</code>
<code>scope_bonus</code>	<code>+0.08</code>	Schema scope matches cue scope exactly
<code>scope_kind_bonus</code>	<code>+0.04</code>	Schema scope kind matches cue scope kind
<code>salience_bonus</code>	<code>+0.02</code>	Schema salience $\geq$ median
<code>utility_bonus</code>	<code>+0.03</code>	Schema utility score $\geq$ median

Identity bonus cap: $B_{\text{identity}} \leq 0.15$ (prevents query-invariant ranking).

Noise penalty P_{noise} :

$$P_{\text{noise}} = w_n \cdot \text{noise_score}(s)$$

Where:

- $w_n = 0.30$ (`_NOISE_PENALTY_WEIGHT`)
- `noise_score(s)` is derived from `shown_count` , `used_count` , `irrelevant_count` feedback history

Step 2: Keyword Scoring (Fallback)

For schemas without embeddings, a simple keyword overlap score is used:

$$\text{score}_{\text{kw}}(s, c) = \frac{|\text{terms}(c) \cap \text{terms}(s)|}{\max(|\text{terms}(c)|, 1)}$$

Step 3: Activation Ranking

Schemas are ranked by descending score. The "rendered" boolean controls whether peripheral schemas are included.

Step 4: MMR Deduplication

Near-duplicates are removed from the ranked list:

$$\text{keep } s_i \text{ iff } \forall j < i : \cos(\mathbf{e}_{s_i}, \mathbf{e}_{s_j}) < \theta_{\text{mmr}}$$

Where $\theta_{\text{mmr}} = 0.92$ — schemas with cosine ≥ 0.92 are considered duplicates. Schemas without embeddings are always kept.

Step 5: Budget Trimming

Two constraints are applied:

1. **Item limit:** at most `max_items` schemas
2. **Character limit:** total rendered text length $\leq$ `max_chars`

Items are included in rank order until either limit is exceeded.

Step 6: Rendering

Each selected schema is rendered as:

```
- [sch_{id}] (peripheral) {content_text}
```

The (peripheral) marker indicates schemas with class/layer outside the explicit allowed sets.

Configuration

GatePolicy

Parameter	Type	Default	Description
max_items	int	10	Maximum number of schemas in context
max_chars	int	4000	Maximum total rendered characters
allowed_classes	tuple[str]	_DEFAULT_ALLOWED_CLASSES	Schema types eligible for identity bonus

_DEFAULT_ALLOWED_CLASSES

("fact", "preference", "interaction_preference", "constraint", "habit", "decision", "lesson"

Excluded Layers

("raw_event", "episodic_summary", "assistant_summary") — these never receive the layer identity bonus.

Excluded Sources

`("assistant_summary", "tool_result_summary")` — these never receive the provenance identity bonus.

Internal Constants

Constant	Symbol	Value	Description
Identity bonus cap	—	0.15	Max query-independent score boost
Noise penalty weight	w_n	0.30	Multiplier for context noise score
MMR cosine threshold	θ_{mmr}	0.92	Near-duplicate detection threshold

Key Invariants

1. Identity bonuses are capped at 0.15 — what a memory IS must only tie-break, never outrank how well it matches the query.
2. Schemas without embeddings fall back to keyword scoring — the system degrades gracefully when the encoder is unavailable.
3. MMR deduplication prevents two near-identical schemas from both occupying token budget.
4. Noise penalty (`_NOISE_PENALTY_WEIGHT = 0.30`) is the primary mechanism for cleaning ranking — it can reduce activation by up to 0.30 compared to salience deltas of ~0.0004.